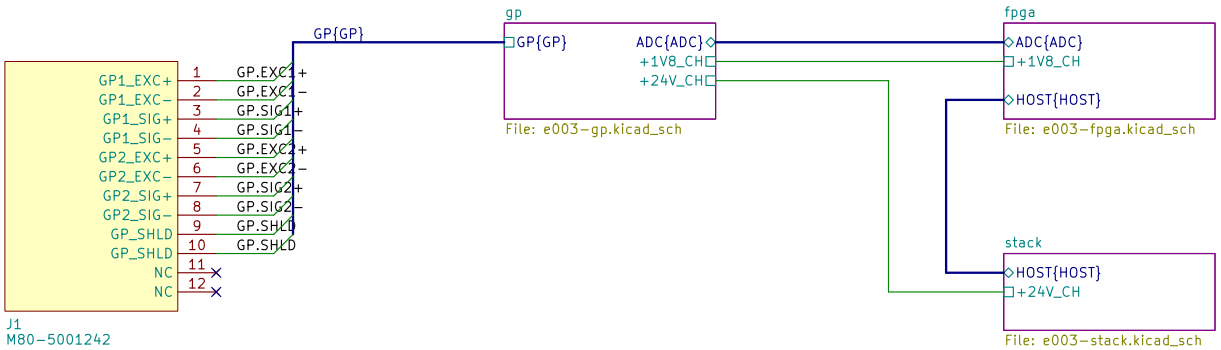


SENSOR CONNECTOR: TWO CHANNELS, A SHIELD PAIR,
PINS 11 AND 12 SPARE

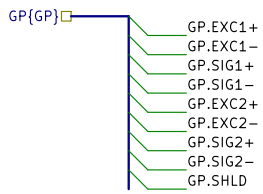
ASRP ADC BOARD: 2 GENERAL-PURPOSE CHANNELS ON THE
STACK'S GROUND, WITH NO ISOLATION. THE FPGA SHEET
DRIVES THE CHANNELS; THE STACK SHEET CARRIES THE
INTERCONNECT, ITS INPUT PROTECTION, THE 3.3 V
CURRENT LIMIT, THE 24 V SENSOR SUPPLY AND THE BUS BUFFER

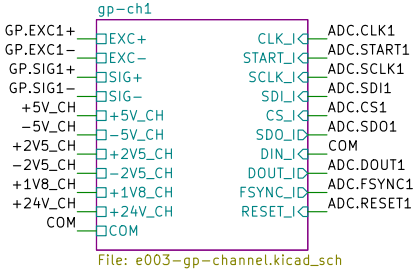


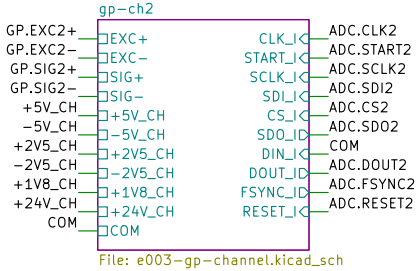
nonplus

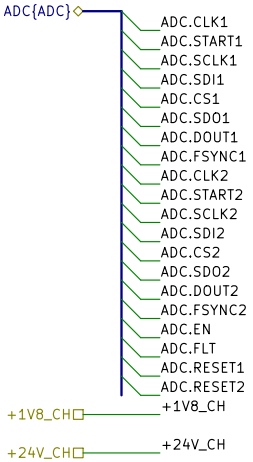
Title: ASRP ADC BOARD

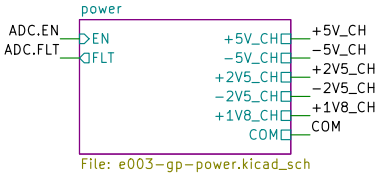
Sheet: /	**Rev: –**
File: e003-adc-board.kicad_sch	Size: A4
	Id: 1/15





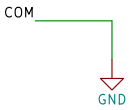




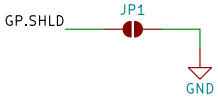


THE TWO CHANNELS SHARE ONE SUPPLY (POWER). EACH HAS ITS OWN REFERENCE AND ITS OWN NINE LINES TO THE FPGA (CLK, START, SCLK, SDI, CS, SDO, DOUT, FSYNC, RESET): NO LINE IS SHARED, SO ONE FAILED CONVERTER CANNOT HOLD THE OTHER'S BUS. DIN, THE DAISY−CHAIN INPUT, IS TIED LOW.

COM, THE CHANNELS' GROUND, IS THE STACK'S GND: THE BOARD HAS ONE DOMAIN AND NO ISOLATION



ADC: THE FPGA'S LINES TO THE CHANNELS, ON ITS TWO 1.8 V BANKS, WITH SUPPLY ENABLE AND FAULT. +1V8_CH POWERS THOSE BANKS. +24V_CH, THE SENSOR SUPPLY, COMES FROM THE STACK THROUGH ITS FUSE (STACK SHEET).



SHIELD: CLOSE THE SOLDER JUMPER WHERE THE SHIELDS ARE TERMINATED AT THE BOARD. IT TIES GP_SHLD TO GND.

nonplus

Title: ASRP ADC BOARD

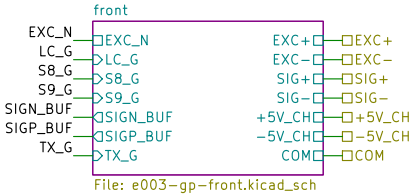
Sheet: /gp/

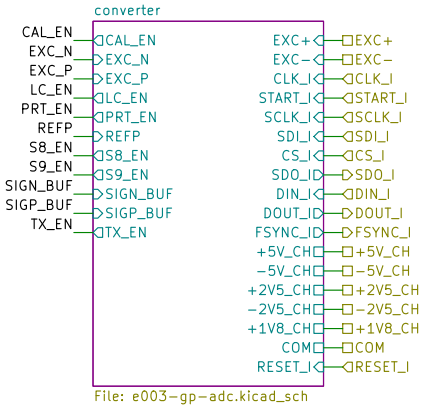
File: e003−gp.kicad_sch

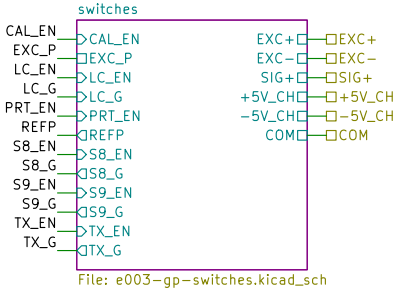
Rev: −

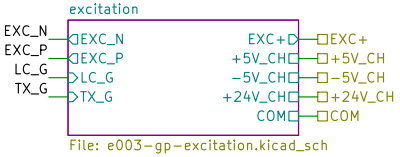
Size: A4

Id: 2/15







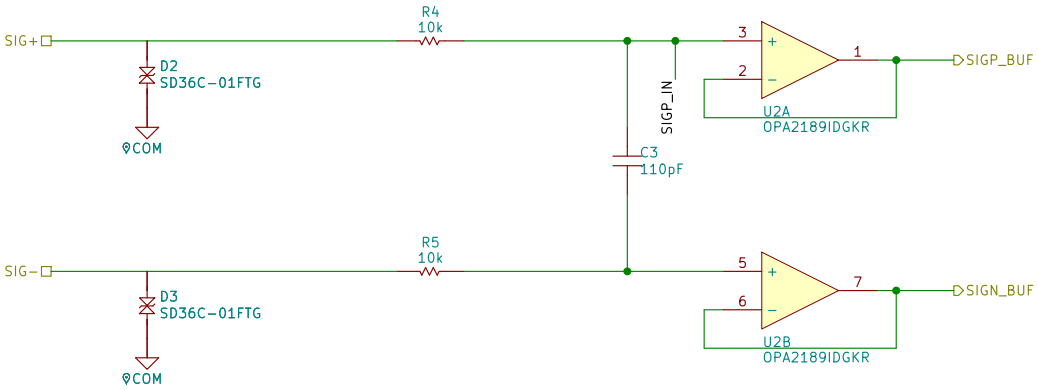


ONE GENERAL−PURPOSE CHANNEL: FRONT END, SWITCHES, EXCITATION AND CONVERTER.
THE TWO CHANNELS ARE INSTANCES OF THIS SHEET. EACH HAS ITS OWN REFERENCE (EXCITATION)
AND ITS OWN LINES TO THE FPGA; THEY SHARE THE BOARD'S RAILS AND COM, WHICH IS THE
STACK'S GND (GP SHEET). EACH CHANNEL'S EXCITATION BUS (EXC_P, EXC_N) STAYS LOCAL

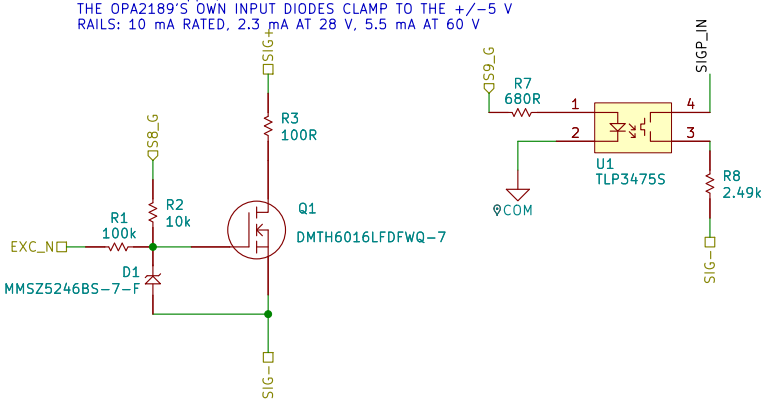
nonplus

Title: ASRP ADC BOARD

Sheet: /gp/gp−ch1/	**Rev: −**
File: e003−gp−channel.kicad_sch	Size: A4
	Id: 3/15

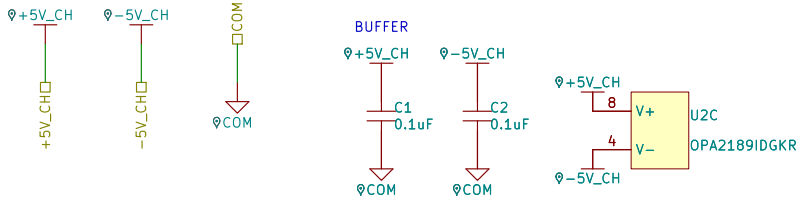


SIGNAL BUFFERS: OPA2189 FROM +/−5 V, 300 pA BIAS
10k SERIES PROTECTION, 0805 FOR A 28 V FAULT;
SIG+ 0.05% 5 ppm/C, ALSO THE TOP OF THE 0–10 V DIVIDER.
THE OPA2189'S OWN INPUT DIODES CLAMP TO THE +/−5 V
RAILS: 10 mA RATED, 2.3 mA AT 28 V, 5.5 mA AT 60 V

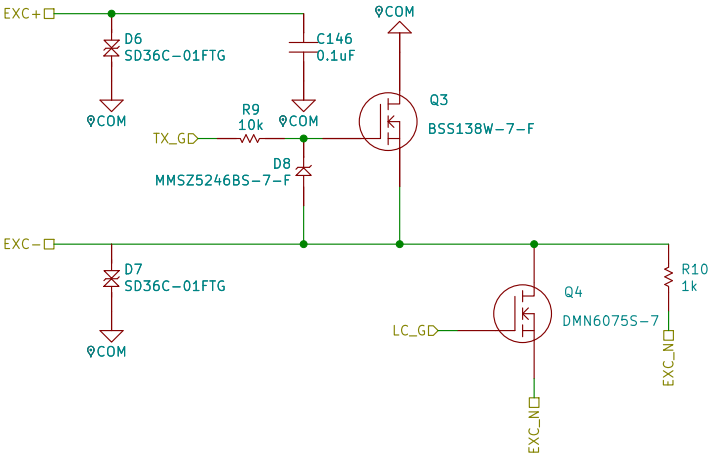


S8: 100R SHUNT ACROSS THE SIGNAL CONTACTS (4–20 mA)
GATE AT EXC_N WHEN OPEN, +5 V CLOSED; 16 V ZENER.
10k IN THE GATE DRIVE (S4 AND S8) LIMITS A CLAMPED FAULT'S
CURRENT INTO THE TMUX6212

S9: PHOTORELAY SWITCHES THE 2.49k DIVIDER BOTTOM
FROM THE SIG+ BUFFER INPUT TO SIG− (0–10 V)
ITS LED RUNS FROM S9_G THROUGH 680R TO COM, ABOUT 5 mA



EXC+: TO THE CURRENT SOURCE, S5 AND THE MONITOR



S4: EXC− TO THE CHANNEL COMMON FOR A TRANSMITTER
S3: EXC− STRAIGHT TO THE BUS FOR A LOAD CELL
1.00k CURRENT-SENSE RESISTOR: A PIEZORESISTIVE BRIDGE'S
REFERENCE; FITTED IN EVERY CONFIGURATION

SUPPRESSORS: SD36C-01FTG, BIDIRECTIONAL, 36 V STAND-OFF,
40 V AT 1 mA, 50 V AT 1 A, 60 V AT 4 A (8/20 us).
S8 IS RATED 60 V. 100 nF ON EXC+ HOLDS A DISCHARGE TO
ABOUT 12 V, WITHIN THE LT3092'S +/−40 V

nonplus

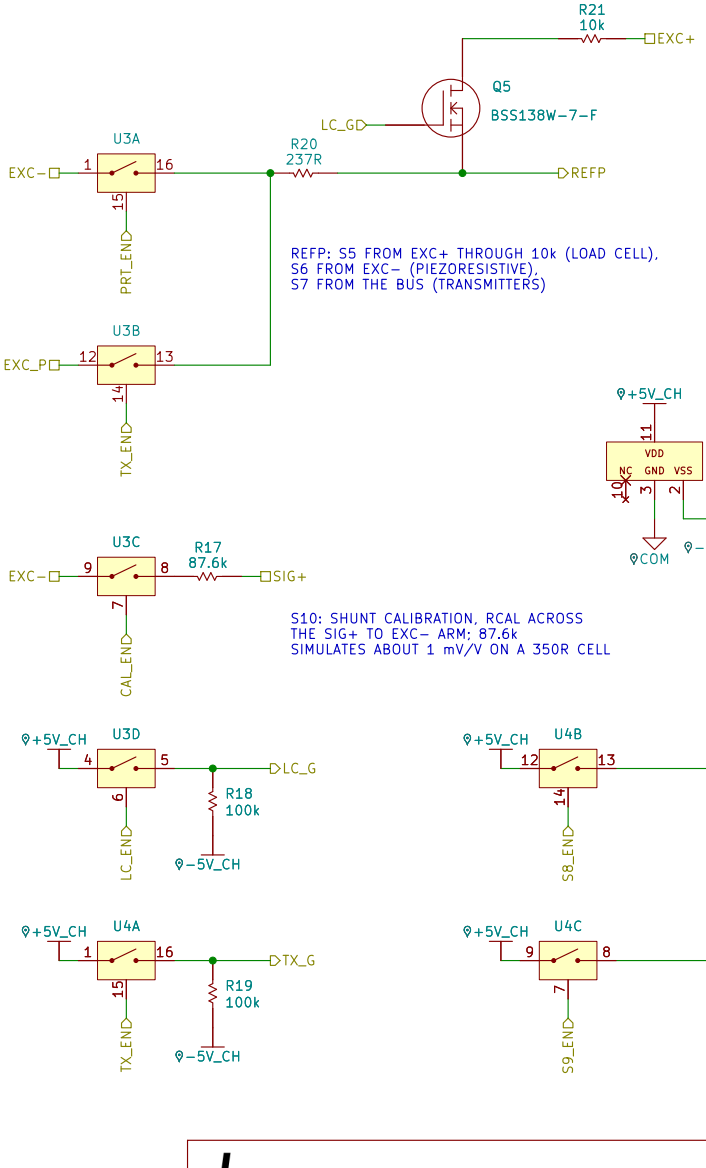
Title: ASRP ADC BOARD

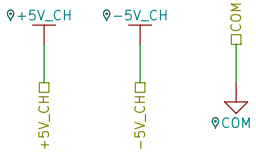
Sheet: /gp/gp-ch1/front/
File: e003-gp-front.kicad_sch

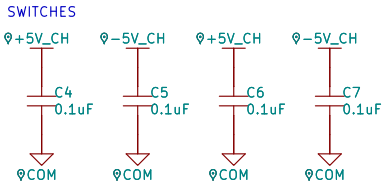
Rev: –
Size: A4
Id: 4/15

SWITCH ENABLES: GPIO0..GPIO5 OF THE CONVERTER, ACTIVE HIGH.
LC: S1, S3, S5 (LOAD CELL) PRT: S6 (PIEZORESISTIVE)
TX: S2, S4, S7 (TRANSMITTER) CAL: S10 (SHUNT CALIBRATION)
S8: 4–20 mA SHUNT S9: VOLTAGE DIVIDER
THE TMUX6212'S OWN 4M PULL-DOWNS: AN UNDRIVEN LINE READS OFF.
NO HARDWARE INTERLOCK: THE FPGA WRITES ONLY THE
COMBINATIONS OF THE CONFIGURATION TABLE (GW-2)

GATE AND LED DRIVE: A CLOSED SWITCH TAKES ITS LINE TO +5 V.
LC_G AND TX_G PULL TO −5 V, AND ALSO LIGHT THE LEDS OF
S1'S AND S2'S PHOTORELAYS (EXCITATION). S8_G IS PULLED
DOWN AT ITS GATE; S9_G LIGHTS S9'S LED THROUGH 680R
TO COM (FRONT END)





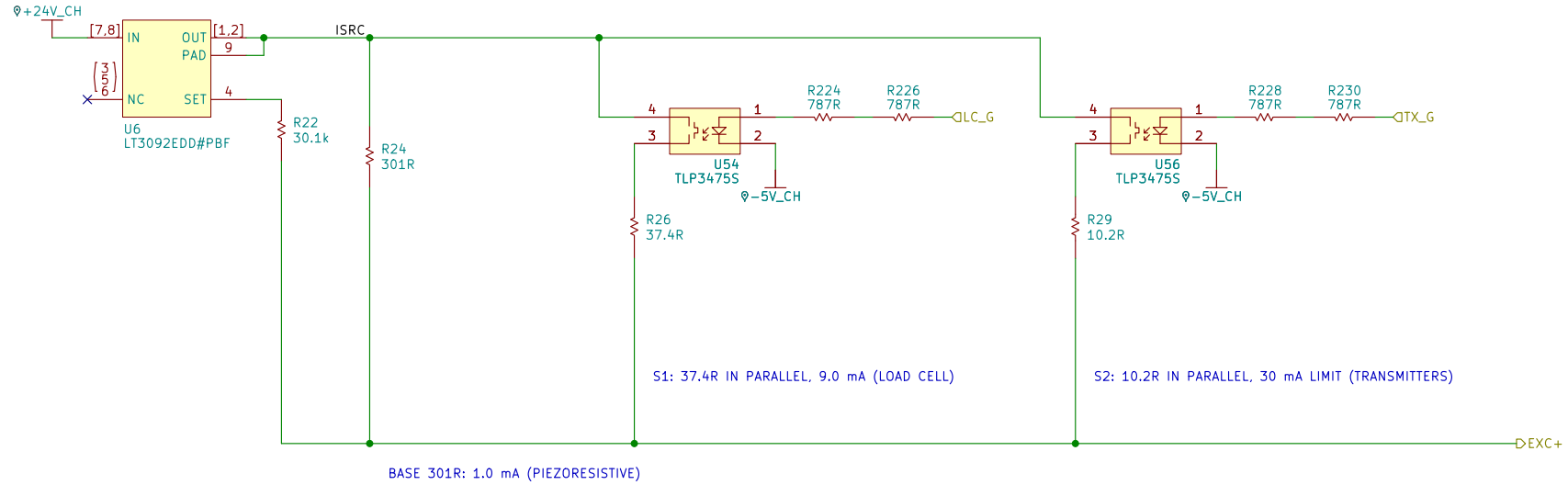


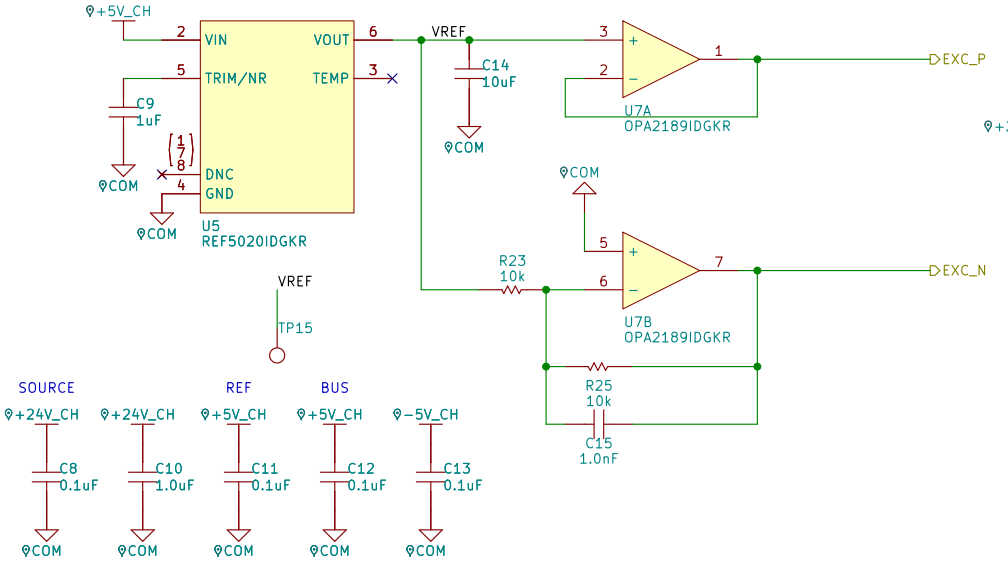
nonplus

Title: ASRP ADC BOARD

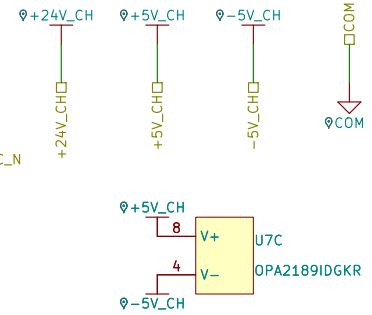
Sheet: /gp/gp-ch1/switches/	Rev: –
File: e003-gp-switches.kicad_sch	Size: A4
	Id: 5/15

SENSOR CURRENT = 10 uA x 30.1k / R(OUT TO EXC+); THE LT3092 HOLDS 0.3 V ACROSS IT.
S1 AND S2: TLP3475S PHOTORELAYS, FLOATING WITH EXC+ (UP TO 24 V), UP TO 1.5 OHM IN THEIR BRANCH. EACH LED RUNS FROM LC_G OR TX_G TO −5 V THROUGH 2 x 787R, ABOUT 5 mA WHEN CLOSED





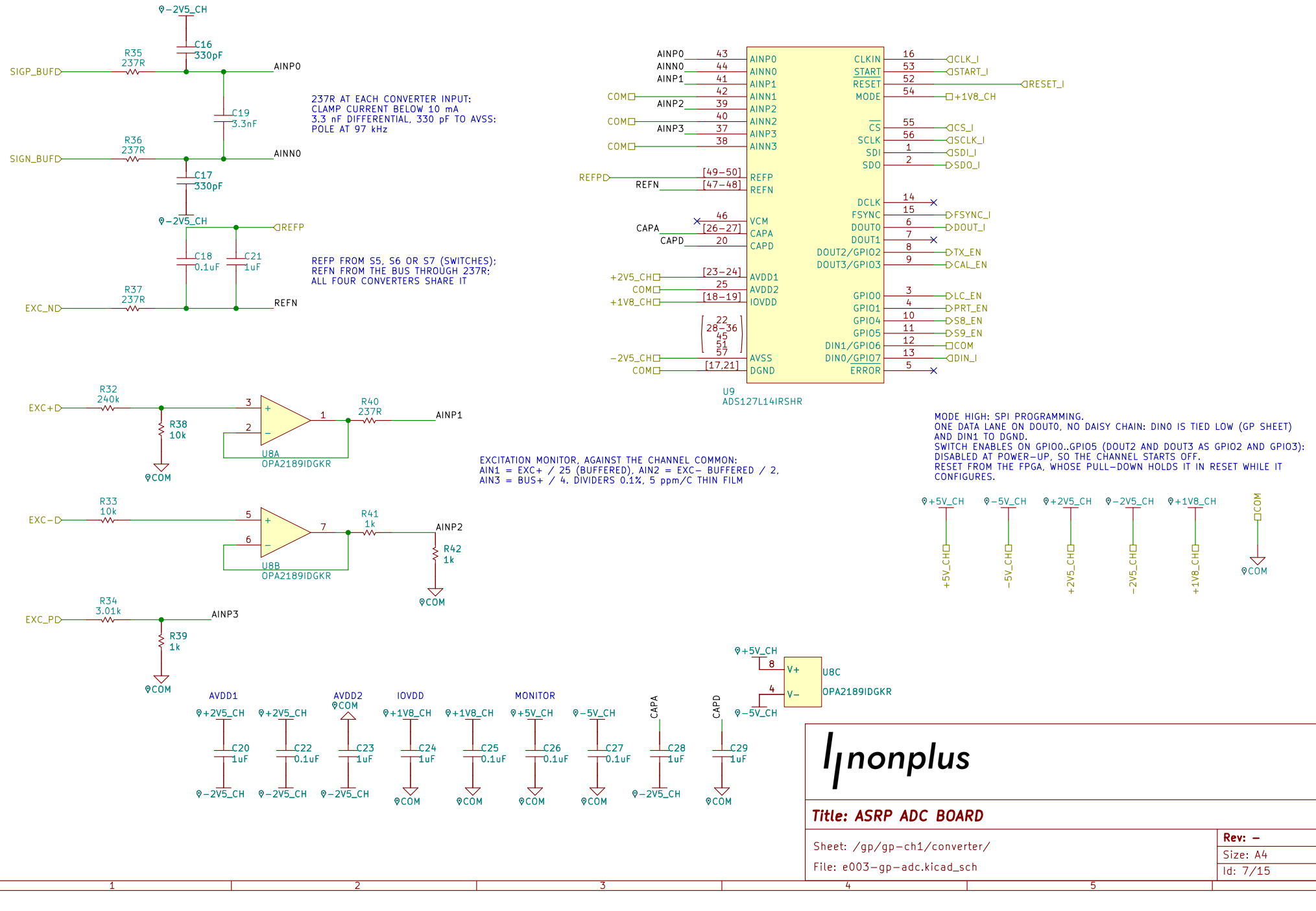
EXCITATION BUS +/−2.048 V: REF5020; ONE HALF OF AN OPA2189 FOLLOWS IT, THE OTHER INVERTS IT, EACH DRIVING ONE SIDE OF THE BUS DIRECTLY.
+/−65 mA OUTPUT LIMIT BOUNDS A SHORT ON THE BUS. INVERTER PAIR 0.05%, 5 ppm/C.
BUS STABILITY WITH THE REFERENCE CAPACITORS TO BE CONFIRMED (OI−8)

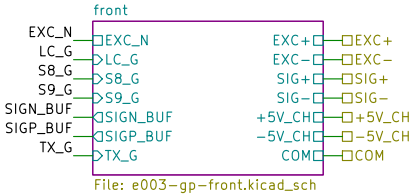


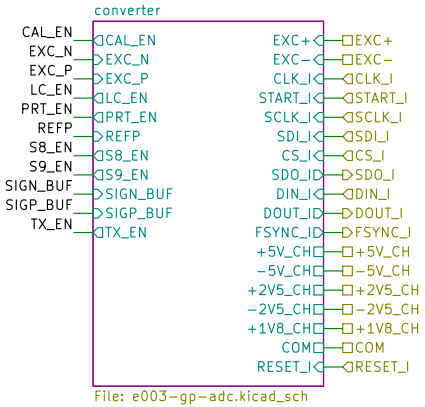
nonplus

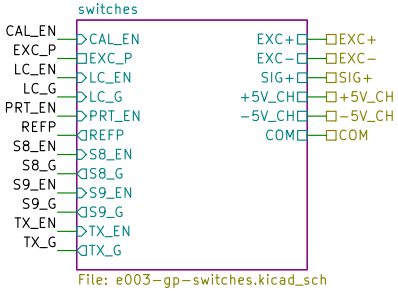
Title: ASRP ADC BOARD

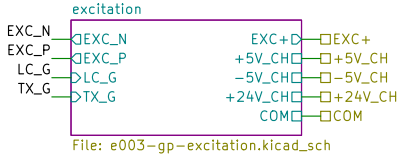
Sheet: /gp/gp−ch1/excitation/	Rev: −
File: e003−gp−excitation.kicad_sch	Size: A4
	Id: 6/15


SIGP_BUF
SIGN_BUF
EXC_N
R35 237R
R36 237R
R37 237R
C16 330pF
C17 330pF
C18 0.1uF
C21 1uF
C19 3.3nF
AINP0
AINN0
REFP
REFN
237R AT EACH CONVERTER INPUT:
CLAMP CURRENT BELOW 10 mA
3.3 nF DIFFERENTIAL, 330 pF TO AVSS:
POLE AT 97 kHz
REFP FROM S5, S6 OR S7 (SWITCHES);
REFN FROM THE BUS THROUGH 237R;
ALL FOUR CONVERTERS SHARE IT
U9
ADS127L14IRSHR
CLKIN
START
RESET
MODE
CS
SCLK
SDI
SDO
DCLK
FSYNC
DOUT0
DOUT1
DOUT2/GPIO2
DOUT3/GPIO3
GPIO0
GPIO1
GPIO4
GPIO5
DIN1/GPIO6
DIN0/GPIO7
ERROR
VCM
CAPA
CAPD
AVDD1
AVDD2
IOVDD
AVSS
DGND
REFP
REFN
AINP0
AINN0
AINP1
AINN1
AINP2
AINN2
AINP3
AINN3
CLK_I
START_I
RESET_I
+1V8_CH
CS_I
SCLK_I
SDI_I
SDO_I
FSYNC_I
DOUT_I
TX_EN
CAL_EN
LC_EN
PRT_EN
S8_EN
S9_EN
COM
DIN_I
MODE HIGH: SPI PROGRAMMING.
ONE DATA LANE ON DOUT0, NO DAISY CHAIN: DIN0 IS TIED LOW (GP SHEET)
AND DIN1 TO DGND.
SWITCH ENABLES ON GPIO0..GPIO5 (DOUT2 AND DOUT3 AS GPIO2 AND GPIO3):
DISABLED AT POWER-UP, SO THE CHANNEL STARTS OFF.
RESET FROM THE FPGA, WHOSE PULL-DOWN HOLDS IT IN RESET WHILE IT
CONFIGURES.
EXC+
EXC-
EXC_P
R32 240k
R38 10k
R33 10k
R34 3.01k
R39 1k
R40 237R
R41 1k
R42 1k
U8A
OPA2189IDGKR
U8B
OPA2189IDGKR
AINP1
AINP2
AINP3
EXCITATION MONITOR, AGAINST THE CHANNEL COMMON:
AIN1 = EXC+ / 25 (BUFFERED), AIN2 = EXC- BUFFERED / 2,
AIN3 = BUS+ / 4. DIVIDERS 0.1%, 5 ppm/C THIN FILM
U8C
OPA2189IDGKR
AVDD1
AVDD2
IOVDD
MONITOR
CAPA
CAPD
C20 1uF
C22 0.1uF
C23 1uF
C24 1uF
C25 0.1uF
C26 0.1uF
C27 0.1uF
C28 1uF
C29 1uF
+5V_CH
-5V_CH
+2V5_CH
-2V5_CH
+1V8_CH
COM
nonplus
Title: ASRP ADC BOARD
Sheet: /gp/gp-ch1/converter/
File: e003-gp-adc.kicad_sch
Rev: -
Size: A4
Id: 7/15







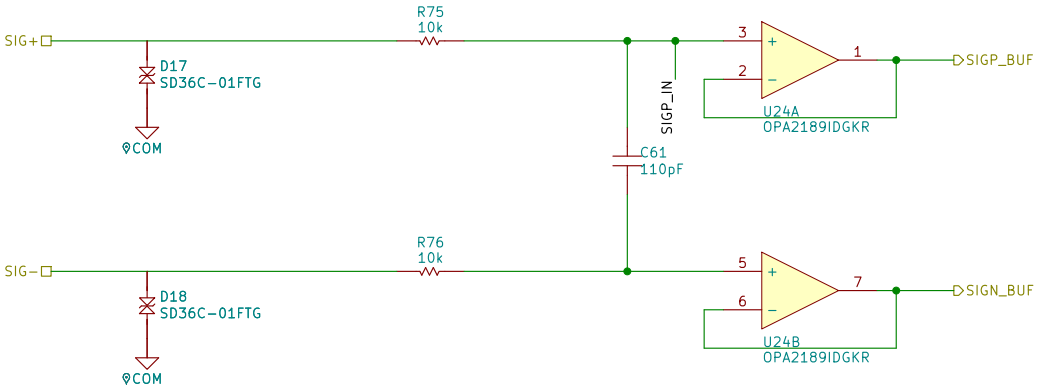


ONE GENERAL−PURPOSE CHANNEL: FRONT END, SWITCHES, EXCITATION AND CONVERTER.
THE TWO CHANNELS ARE INSTANCES OF THIS SHEET. EACH HAS ITS OWN REFERENCE (EXCITATION)
AND ITS OWN LINES TO THE FPGA; THEY SHARE THE BOARD'S RAILS AND COM, WHICH IS THE
STACK'S GND (GP SHEET). EACH CHANNEL'S EXCITATION BUS (EXC_P, EXC_N) STAYS LOCAL

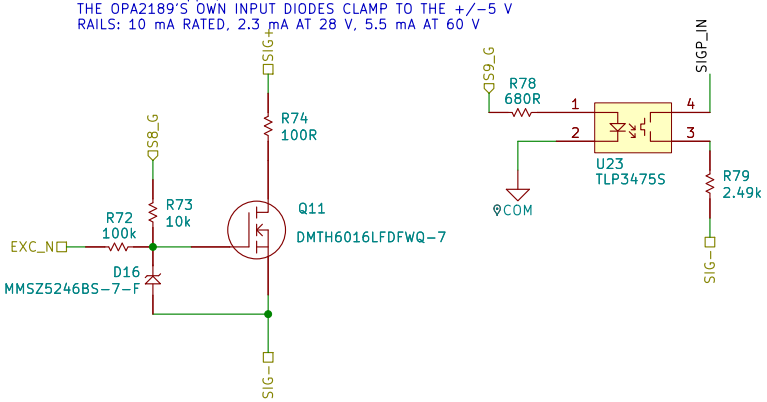
nonplus

Title: ASRP ADC BOARD

Sheet: /gp/gp−ch2/	**Rev: −**
File: e003−gp−channel.kicad_sch	Size: A4
	Id: 8/15

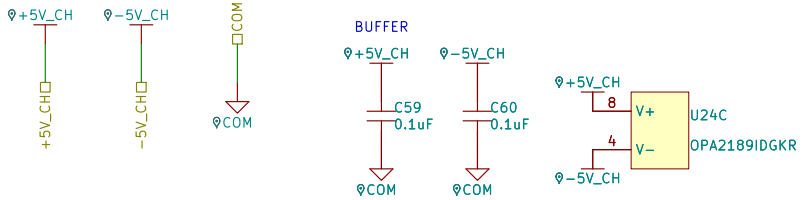


SIGNAL BUFFERS: OPA2189 FROM +/−5 V, 300 pA BIAS
10k SERIES PROTECTION, 0805 FOR A 28 V FAULT;
SIG+ 0.05% 5 ppm/C, ALSO THE TOP OF THE 0–10 V DIVIDER.
THE OPA2189'S OWN INPUT DIODES CLAMP TO THE +/−5 V
RAILS: 10 mA RATED, 2.3 mA AT 28 V, 5.5 mA AT 60 V

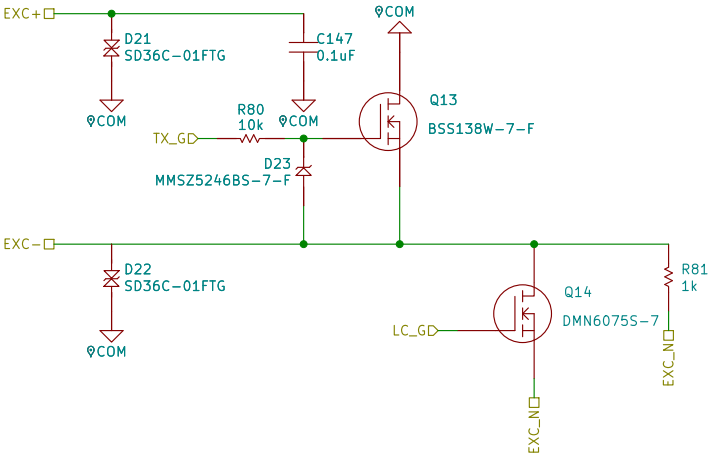


S8: 100R SHUNT ACROSS THE SIGNAL CONTACTS (4–20 mA)
GATE AT EXC_N WHEN OPEN, +5 V CLOSED; 16 V ZENER.
10k IN THE GATE DRIVE (S4 AND S8) LIMITS A CLAMPED FAULT'S

S9: PHOTORELAY SWITCHES THE 2.49k DIVIDER BOTTOM
FROM THE SIG+ BUFFER INPUT TO SIG− (0–10 V)
ITS LED RUNS FROM S9_G THROUGH 680R TO COM, ABOUT 5 mA
CURRENT INTO THE TMUX6212



EXC+: TO THE CURRENT SOURCE, S5 AND THE MONITOR



S4: EXC− TO THE CHANNEL COMMON FOR A TRANSMITTER
S3: EXC− STRAIGHT TO THE BUS FOR A LOAD CELL
1.00k CURRENT-SENSE RESISTOR: A PIEZORESISTIVE BRIDGE'S
REFERENCE; FITTED IN EVERY CONFIGURATION

SUPPRESSORS: SD36C-01FTG, BIDIRECTIONAL, 36 V STAND-OFF,
40 V AT 1 mA, 50 V AT 1 A, 60 V AT 4 A (8/20 us).
S8 IS RATED 60 V. 100 nF ON EXC+ HOLDS A DISCHARGE TO
ABOUT 12 V, WITHIN THE LT3092'S +/−40 V

nonplus

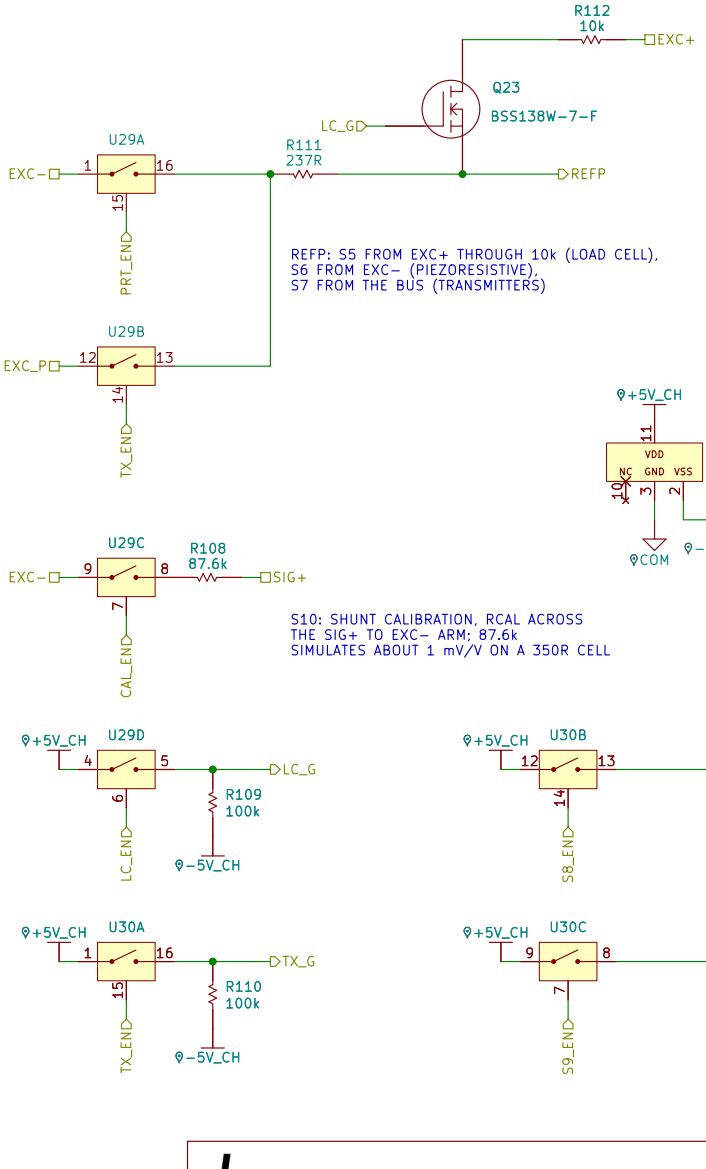
Title: ASRP ADC BOARD

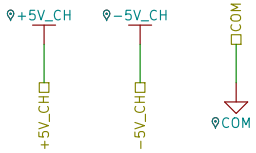
Sheet: /gp/gp-ch2/front/
File: e003-gp-front.kicad_sch

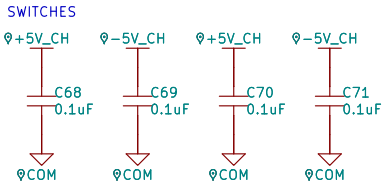
Rev: –
Size: A4
Id: 9/15

SWITCH ENABLES: GPIO0..GPIO5 OF THE CONVERTER, ACTIVE HIGH.
LC: S1, S3, S5 (LOAD CELL) PRT: S6 (PIEZORESISTIVE)
TX: S2, S4, S7 (TRANSMITTER) CAL: S10 (SHUNT CALIBRATION)
S8: 4–20 mA SHUNT S9: VOLTAGE DIVIDER
THE TMUX6212'S OWN 4M PULL-DOWNS: AN UNDRIVEN LINE READS OFF.
NO HARDWARE INTERLOCK: THE FPGA WRITES ONLY THE
COMBINATIONS OF THE CONFIGURATION TABLE (GW-2)

GATE AND LED DRIVE: A CLOSED SWITCH TAKES ITS LINE TO +5 V.
LC_G AND TX_G PULL TO −5 V, AND ALSO LIGHT THE LEDS OF
S1'S AND S2'S PHOTORELAYS (EXCITATION). S8_G IS PULLED
DOWN AT ITS GATE; S9_G LIGHTS S9'S LED THROUGH 680R
TO COM (FRONT END)







nonplus

Title: ASRP ADC BOARD

Sheet: /gp/gp-ch2/switches/

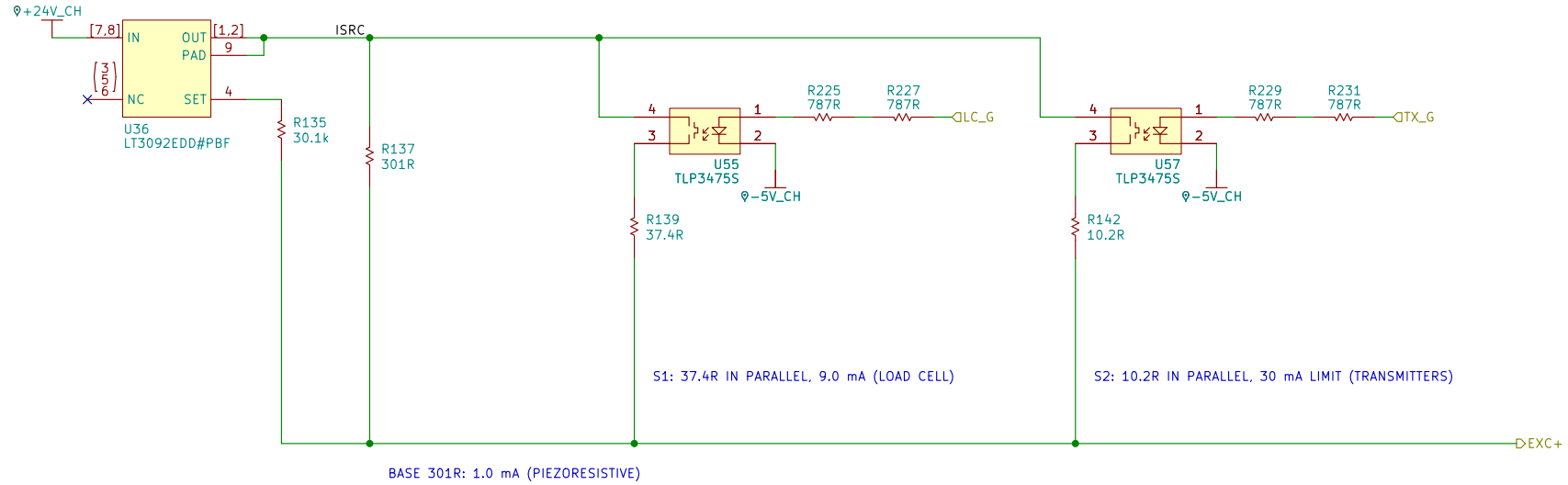
File: e003-gp-switches.kicad_sch

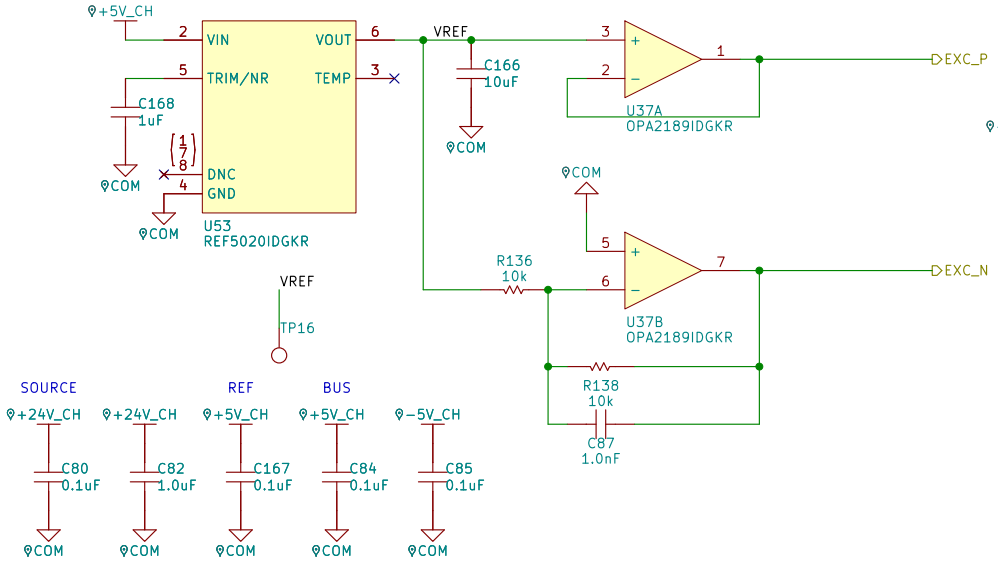
Rev: –

Size: A4

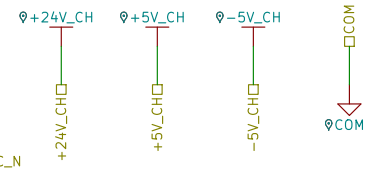
Id: 10/15

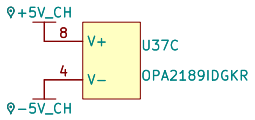
SENSOR CURRENT = 10 uA x 30.1k / R(OUT TO EXC+); THE LT3092 HOLDS 0.3 V ACROSS IT.
S1 AND S2: TLP3475S PHOTORELAYS, FLOATING WITH EXC+ (UP TO 24 V), UP TO 1.5 OHM IN THEIR BRANCH. EACH LED RUNS FROM LC_G OR TX_G TO −5 V THROUGH 2 x 787R, ABOUT 5 mA WHEN CLOSED





EXCITATION BUS +/−2.048 V: REF5020; ONE HALF OF AN OPA2189 FOLLOWS IT, THE OTHER INVERTS IT, EACH DRIVING ONE SIDE OF THE BUS DIRECTLY.
+/−65 mA OUTPUT LIMIT BOUNDS A SHORT ON THE BUS. INVERTER PAIR 0.05%, 5 ppm/C.
BUS STABILITY WITH THE REFERENCE CAPACITORS TO BE CONFIRMED (OI−8)



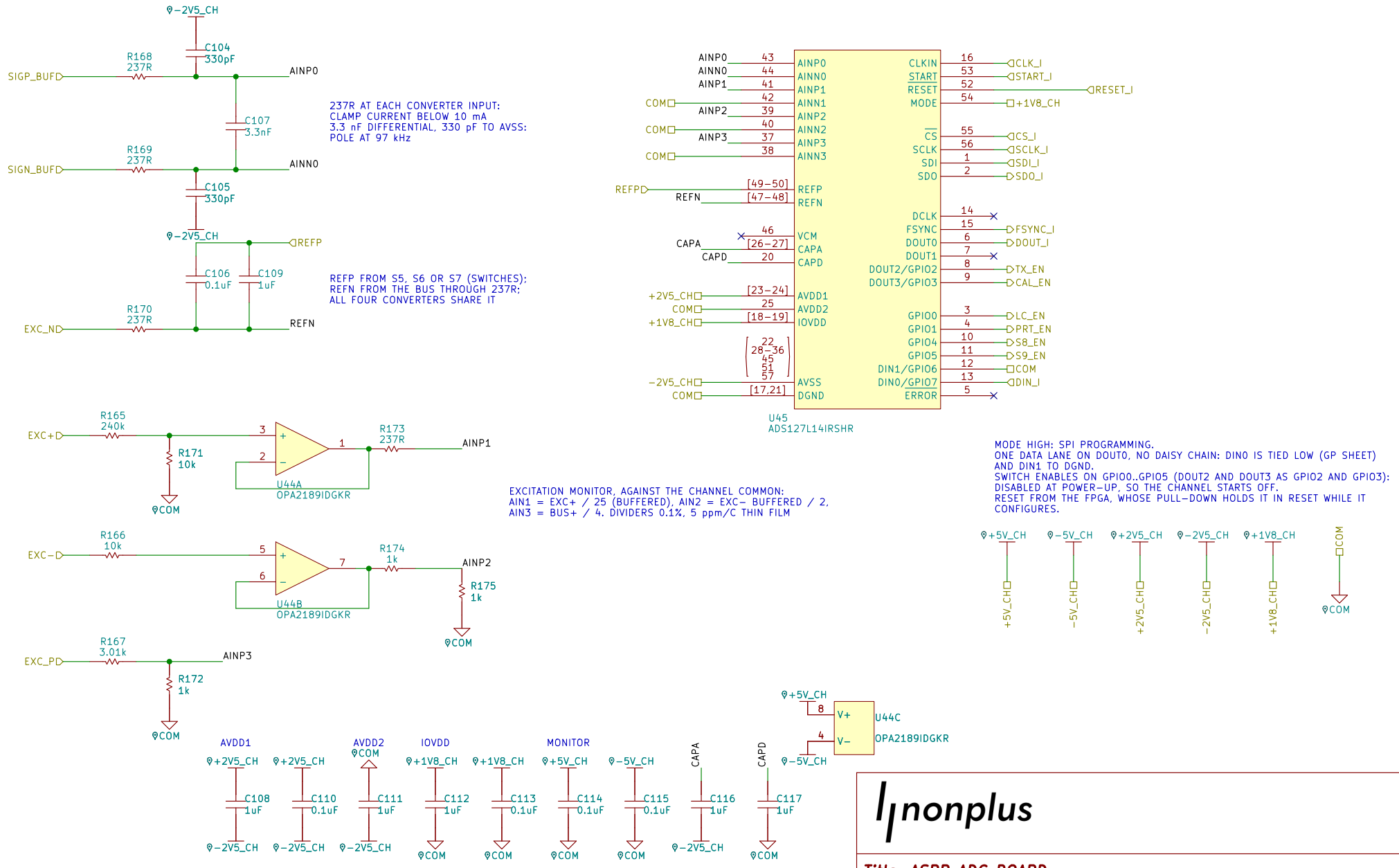


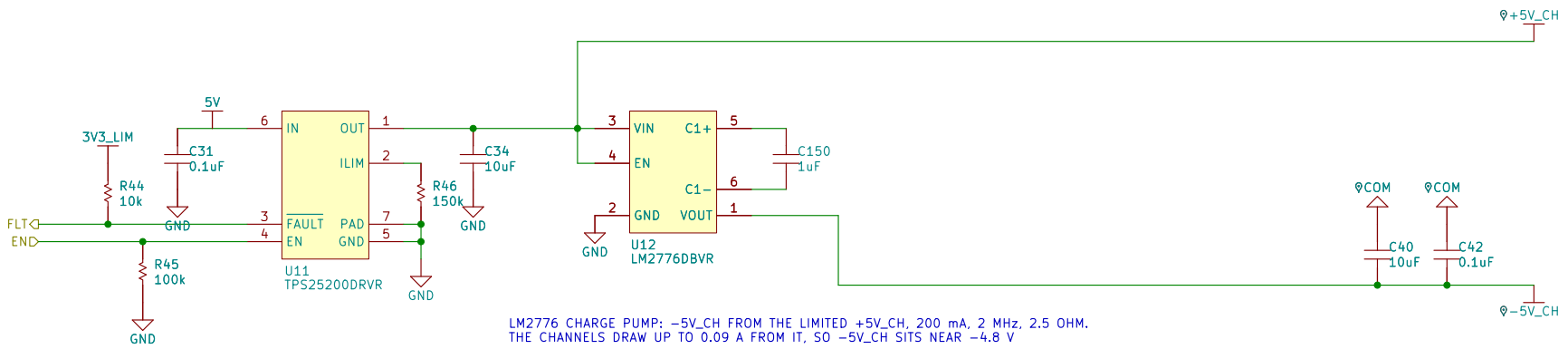
nonplus

Title: ASRP ADC BOARD

Sheet: /gp/gp−ch2/excitation/
File: e003−gp−excitation.kicad_sch

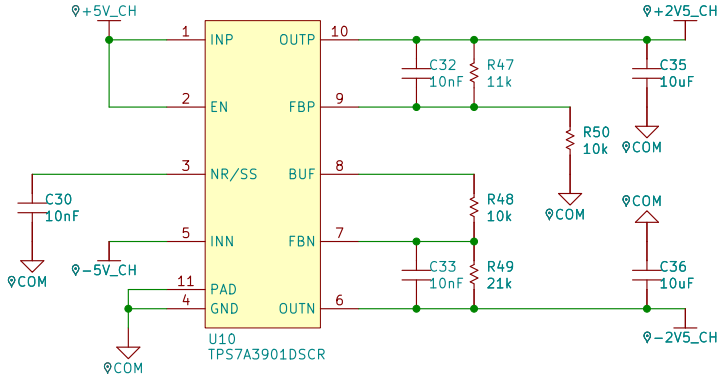
Rev: −
Size: A4
Id: 11/15


237R AT EACH CONVERTER INPUT:
CLAMP CURRENT BELOW 10 mA
3.3 nF DIFFERENTIAL, 330 pF TO AVSS:
POLE AT 97 kHz
REFP FROM S5, S6 OR S7 (SWITCHES);
REFN FROM THE BUS THROUGH 237R;
ALL FOUR CONVERTERS SHARE IT
EXCITATION MONITOR, AGAINST THE CHANNEL COMMON:
AIN1 = EXC+ / 25 (BUFFERED), AIN2 = EXC− BUFFERED / 2,
AIN3 = BUS+ / 4. DIVIDERS 0.1%, 5 ppm/C THIN FILM
MODE HIGH: SPI PROGRAMMING.
ONE DATA LANE ON DOUT0, NO DAISY CHAIN: DIN0 IS TIED LOW (GP SHEET)
AND DIN1 TO DGND.
SWITCH ENABLES ON GPIO0..GPIO5 (DOUT2 AND DOUT3 AS GPIO2 AND GPIO3):
DISABLED AT POWER−UP, SO THE CHANNEL STARTS OFF.
RESET FROM THE FPGA, WHOSE PULL−DOWN HOLDS IT IN RESET WHILE IT
CONFIGURES.
U45
ADS127L14IRSHR
U44A
OPA2189IDGKR
U44B
OPA2189IDGKR
U44C
OPA2189IDGKR
nonplus
Title: ASRP ADC BOARD
Sheet: /gp/gp−ch2/converter/
File: e003−gp−adc.kicad_sch
Rev: −
Size: A4
Id: 12/15

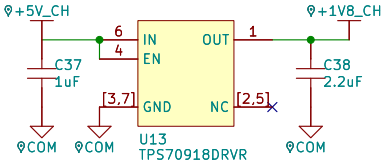


CHANNEL SUPPLY LIMIT: TPS25200 eFUSE AT 0.65 A (150k, 0.59 TO 0.71 A).
+5V_CH IS THE STACK'S 5 V; THE FPGA SWITCHES IT AND READS THE FAULT FLAG.
IN WITHSTANDS 20 V, ABOVE THE TVS0500'S 9.2 V CLAMP (STACK SHEET);
OUT CLAMPS AT 5.4 V, AND IN ABOVE 7.6 V TURNS IT OFF

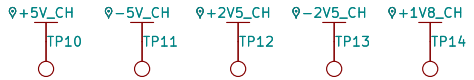
LM2776 CHARGE PUMP: −5V_CH FROM THE LIMITED +5V_CH, 200 mA, 2 MHz, 2.5 OHM.
THE CHANNELS DRAW UP TO 0.09 A FROM IT, SO −5V_CH SITS NEAR −4.8 V



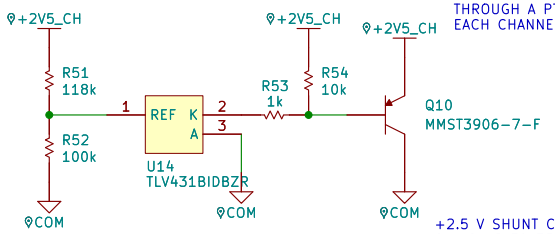
TPS7A39: +/−2.5 V, OUTPUTS RISE TOGETHER
VOUT+ = 1.2 V (1 + 11.0k / 10k)
VOUT− = −1.2 V x 21.0k / 10k



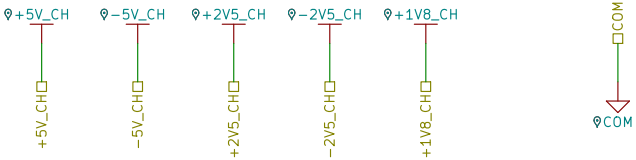
RAIL TEST POINTS



24 V SENSOR SUPPLY: +24V_CH COMES FROM THE STACK'S 24 V
THROUGH A PTC FUSE AND A TVS2700 (STACK SHEET), INTO
EACH CHANNEL'S LT3092 (EXCITATION SHEETS)



+2.5 V SHUNT CLAMP AT 2.7 V = 1.24 V (1 + 118k / 100k):
SINKS UP TO 35 mA, ENOUGH FOR THE FOUR SIGNAL BUFFERS OF
BOTH CHANNELS DRIVEN PAST THEIR RANGE AT ONCE (8 mA EACH
INTO THE CONVERTERS' CLAMPS). DRAWS NOTHING IN NORMAL OPERATION



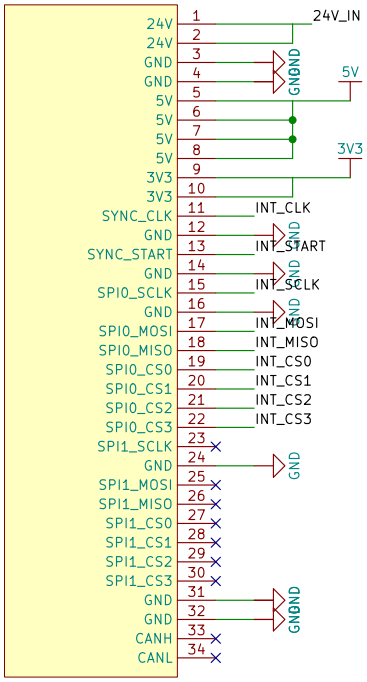
nonplus

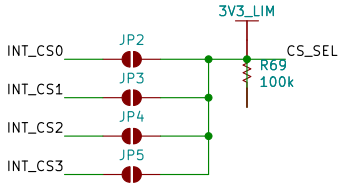
Title: ASRP ADC BOARD

Sheet: /gp/power/
File: e003−gp−power.kicad_sch

Rev: −
Size: A4
Id: 13/15

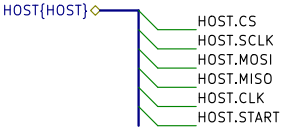
ASRP-PCB-INTERCONNECT-2: 34-WAY PASS-THROUGH (PART TBD).
SPI1 SERVES THE E004-TC-BOARD AND CANH/CANL THE CAN FD BOARDS; BOTH PASS THROUGH UNUSED

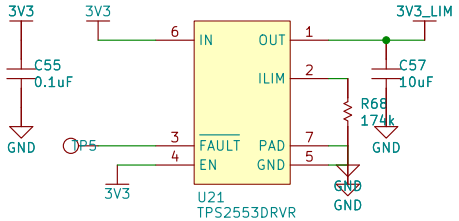




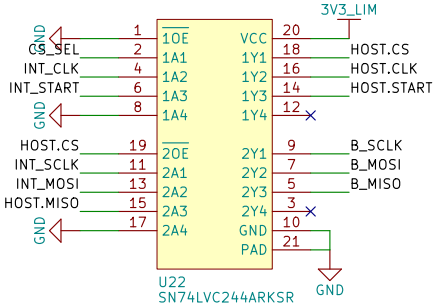
CHIP SELECT: CLOSE ONE SOLDER JUMPER TO PICK SPI0_CS0
TO CS3; 100k HOLDS A BOARD WITH NONE CLOSED DESELECTED

HOST: THE BUFFERED SPI, SYNC_CLK AND SYNC_START
TO THE FPGA

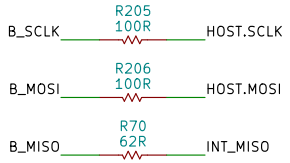




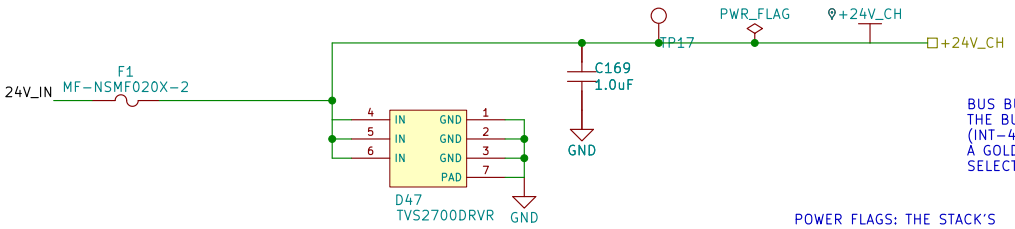
INPUT CURRENT LIMIT: 3.3 V AT 0.15 A (174k), FAULT FLAG TO TP5.
THE 5 V GOES STRAIGHT TO THE CHANNEL SUPPLY LIMIT (GP POWER SHEET),
SO A BOARD THAT FAILS SHORT IS HELD OFF BOTH STACK RAILS

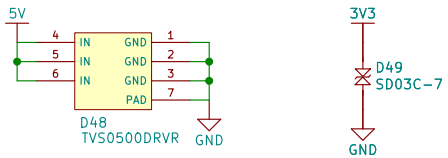


SERIES RESISTORS: 100R ON SCLK AND MOSI TO THE FPGA,
62R ON MISO TO THE INTERCONNECT

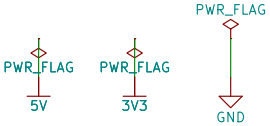


INPUT PROTECTION: TVS0500 ON 5 V (8.4 V MAX. BREAKDOWN, 9.2 V AT 43 A,
WITHIN THE 20 V OF THE CHANNEL LIMIT'S INPUT, GP POWER SHEET);
SD03C ON 3.3 V (3.3 V STAND-OFF, 4.0 V MIN. BREAKDOWN, 5.0 V AT 1 A, 5.7 V AT 10 A);
24 V THROUGH A 0.2 A PTC (MF-NSMF020X) TO A TVS2700
(33.9 V AT 40 A, BELOW THE LT3092'S 40 V), THEN TO +24V_CH

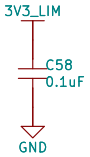




POWER FLAGS: THE STACK'S
RAILS ENTER ON J2



BUS BUFFER, 3.3 V: GROUP 1 ALWAYS ENABLED (CS, SYNC_CLK, SYNC_START).
THE BUFFERED CS ENABLES GROUP 2: SCLK AND MOSI IN, MISO OUT THROUGH 62R
(INT-4). DESELECTED, GROUP 2 RELEASES THE FPGA'S CONFIGURATION SPI PINS, SO
A GOLDEN-IMAGE BOOT CAN DRIVE THEM; 100R LIMITS A CLASH IF THE BOARD IS
SELECTED DURING ONE



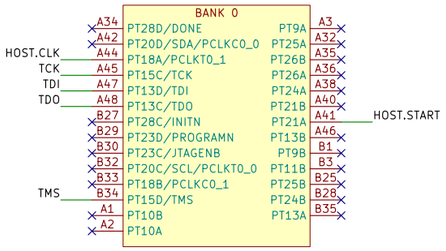
nonplus

Title: ASRP ADC BOARD

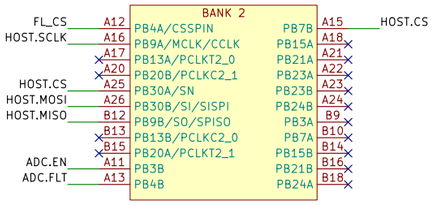
Sheet: /stack/
File: e003-stack.kicad_sch

Rev: -
Size: A4
Id: 14/15

BANKS 1 AND 4 AT 1.8 V (+1V8_CH) FACE THE CONVERTERS, ONE EACH, WITH EACH CONVERTER'S RESET; BANKS 0 AND 2 AT 3.3 V.
BANK 2 HOLDS THE CONFIGURATION SPI PINS: THE HOST'S SLAVE SPI (CCLK, SI, SO, SN) AND THE GOLDEN-IMAGE FLASH'S MASTER SPI (MCLK, SISPI, SPISO, CSSPIN). THE HOST'S CHIP SELECT ALSO GOES TO PB7B, THE USER SPI'S CHIP SELECT (SPI_SCSN)

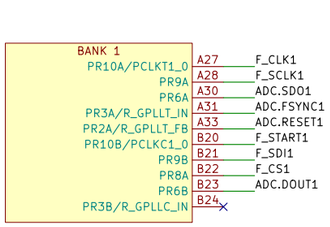


U50A
LCMXO2-4000HC-4QN84C

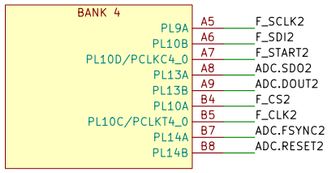


U50C
LCMXO2-4000HC-4QN84C

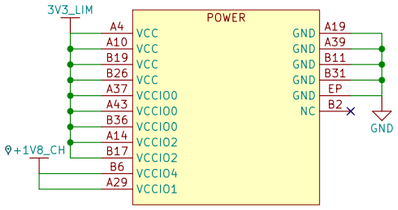
HOST{HOST}
HOST.CS
HOST.SCLK
HOST.MOSI
HOST.MISO
HOST.CLK
HOST.START



U50B
LCMXO2-4000HC-4QN84C

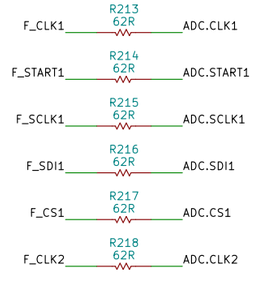


U50D
LCMXO2-4000HC-4QN84C

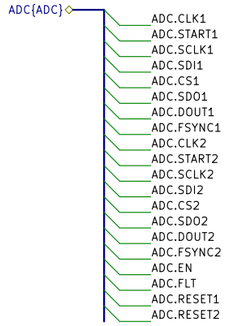


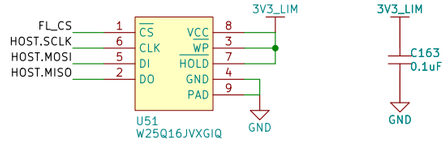
U50E
LCMXO2-4000HC-4QN84C

+1V8_CH
+1V8_CH

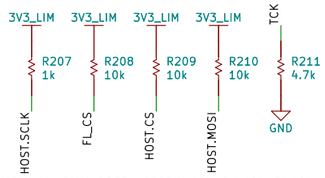


62R SERIES RESISTORS AT THE FPGA'S OUTPUTS TO THE CONVERTERS: CLK, START, SCLK, SDI AND CS OF EACH CHANNEL





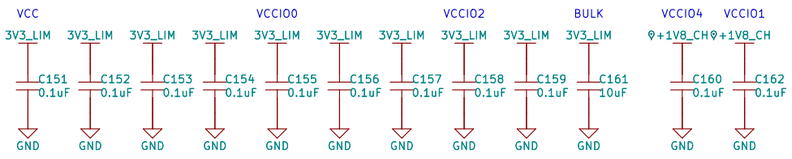
GOLDEN IMAGE: W25Q16JV ON THE MASTER SPI PINS (MCLK, SISPI, SPISO, CSSPIN), SHARED WITH THE SLAVE SPI PORT THAT THE HOST USES; DUAL BOOT LOADS IT ONLY WHEN THE INTERNAL IMAGE FAILS



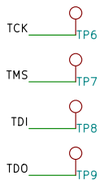
MCLK 1k (DUAL BOOT), CSSPIN 10k, SN 10k, SI 10k
(FPGA-TN-02155): TCK 4.7k DOWN

RECONFIGURATION (GW-9): THE HOST SENDS REFRESH (0x79 00 00) TO THE SLAVE SPI PORT, WHICH EVERY IMAGE KEEPS ENABLED; THE HARDENED CONFIGURATION LOGIC RELOADS THE FPGA WHATEVER THE USER DESIGN IS DOING. PROGRAMN IS OPEN (WEAK PULL-UP IN A BLANK DEVICE)

DECOUPLING: 0.1 uF AT EACH SUPPLY PIN; BANKS 1 AND 4 (VCCIO1, VCCIO4) AT 1.8 V



JTAG TEST POINTS



nonplus

Title: ASRP ADC BOARD

Sheet: /fpga/
File: e003-fpga.kicad_sch

Rev: –
Size: A3
Id: 15/15